

Q2: Percolation in 1-D Lattice

UAS Spring '26 | Roll Number: 203101108

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Parameters

Since $203101108 \bmod 5 = 3$, the lattice sizes used throughout are:

Symbol	Formula	Value	Used In
L_1	$(\text{roll mod } 5) + 2$	5	Spanning probability plot
L_2	1 (fixed)	1	Spanning probability plot
L_3	$(\text{roll mod } 5) + 10$	13	Spanning probability plot
L_4	$((\text{roll mod } 5) + 1) \times 100$	400	Average cluster size plot

Part 1: Spanning Probability $\Pi(p ; L)$

Theory

In 1D site percolation, each site is independently occupied with probability p . A spanning cluster (connecting both ends of the lattice) exists **only if every one of the L sites is simultaneously occupied**, because a single empty site creates an irreparable gap. The spanning probability is therefore:

$$\Pi(p ; L) = p^L$$

This is the probability that L independent Bernoulli(p) events all succeed.

Results

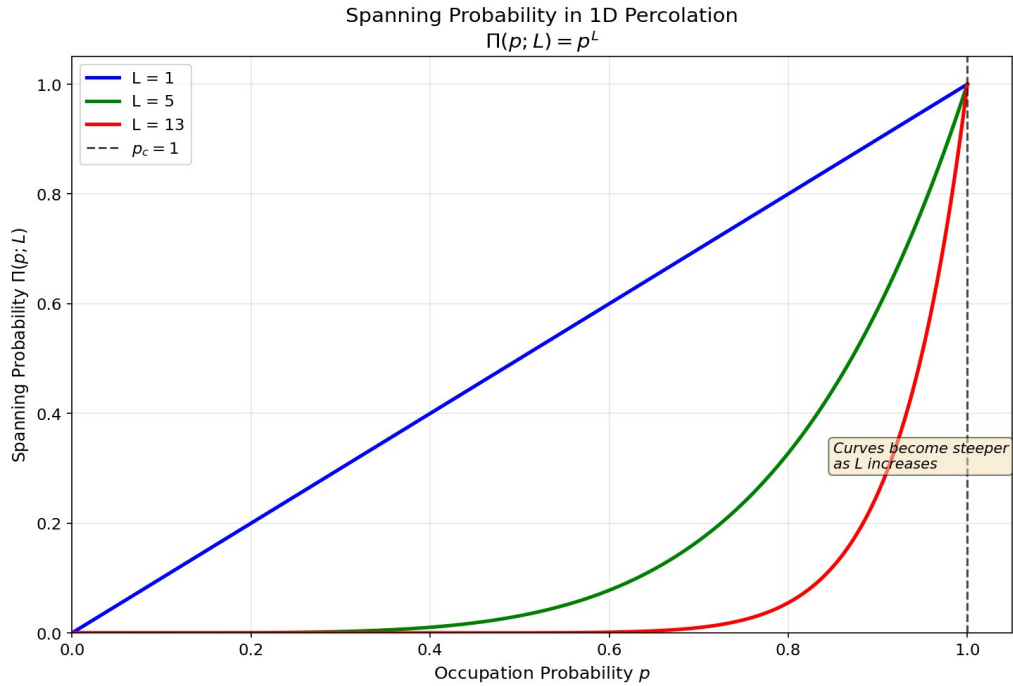


Figure 1: Spanning probability $\Pi(p; L) = p^L$ for $L = 1, 5$, and 13 . The dashed vertical line marks $p_c = 1$.

Interpretation of the Three Curves

Curve	L	Formula	Behaviour and Observations
Blue	1	$\Pi = p$	Linear. Any single occupied site trivially spans a lattice of size 1.
Green	5	$\Pi = p^5$	Moderately concave. Spanning probability is 0.33 at $p = 0.8$. Drops sharply below $p = 0.7$.
Red	13	$\Pi = p^{13}$	Very steep near $p = 1$. Even at $p = 0.9$ the value is only 0.25. Effectively zero for p below 0.8.

Key observation: As L increases the curves become progressively steeper and their inflection points shift toward $p = 1$. In the thermodynamic limit ($L \rightarrow \infty$), the curve p^L converges to a perfect step function at $p = 1$.

Why the Critical Probability is $p_c = 1$

In the thermodynamic limit:

$$\Pi(p; L) = p^L \rightarrow 0 \text{ for any } p < 1 \text{ (as } L \rightarrow \infty)$$

$$\Pi(p; L) = p^L \rightarrow 1 \text{ only when } p = 1$$

Physical reasoning: In 1D there are no alternative paths around an empty site. A spanning cluster therefore requires **every** site to be occupied. Because the probability of this is $p^L \rightarrow 0$ for any $p < 1$ as L grows, genuine percolation occurs only at $p_c = 1$. This contrasts with 2D ($p_c \approx 0.593$ for the square lattice) and 3D lattices, where alternative paths allow percolation at finite p .

Part 2: Average Cluster Size $S(p)$

Theory

The average cluster size (counting only finite clusters) for an infinite 1D lattice is given exactly by:

$$S(p) = (1 + p) / (1 - p) \text{ for } p < 1$$

Derivation sketch: A finite cluster of size s has probability $(1 - p)^2 p^{s-1}$ (two empty boundary sites and s occupied interior sites). Summing $s \times \text{Prob}(\text{cluster of size } s)$ over all s and normalising yields the formula above.

Critical Behaviour Near p_c

As $p \rightarrow p_c = 1$, define $\varepsilon = 1 - p$ (small and positive). Then:

$$S(p) = (1 + p) / (1 - p) \approx 2 / (1 - p) = 2 / \varepsilon = 2 (p_c - p)^{-1}$$

This is a power-law divergence with critical exponent $\gamma = 1$:

$$S(p) \sim (p_c - p)^{-\gamma} \text{ with } \gamma = 1$$

Results ($L = 400$)

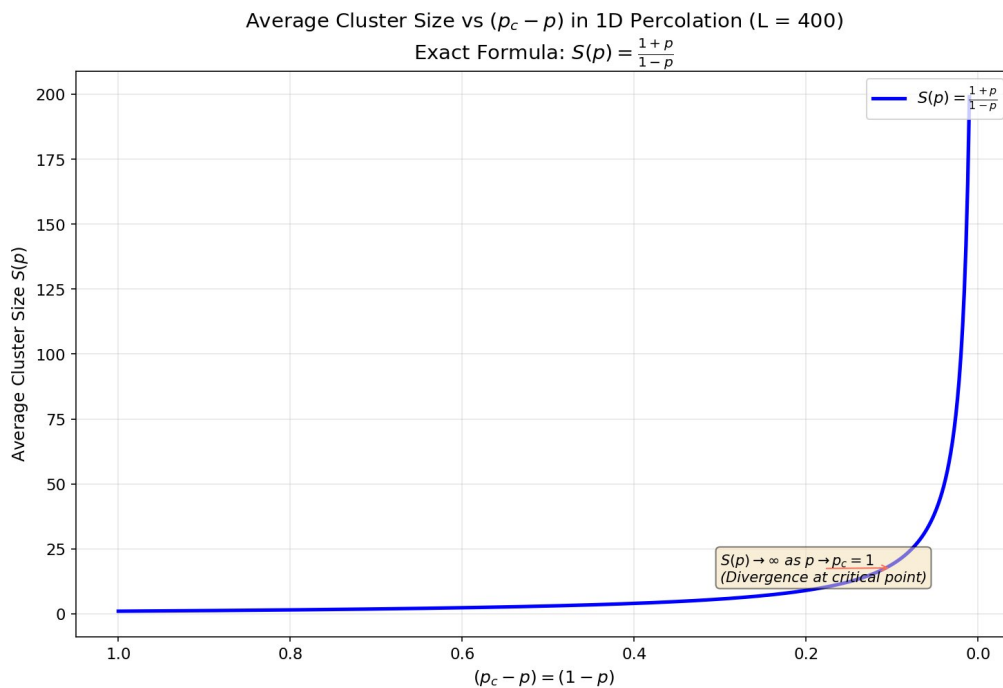


Figure 2: Average cluster size $S(p)$ as a function of $(p_c - p) = (1 - p)$ for $L = 400$. The x-axis is inverted so p increases to the right, approaching $p_c = 1$ at the right edge.

Interpretation

Region	p range	$(p_c - p)$ range	Behaviour
Far below p_c	0 to 0.5	1.0 to 0.5	$S(p) \approx 1-3$. Sites mostly isolated; clusters are tiny.

Region	p range	(p _c - p) range	Behaviour
Intermediate	0.5 to 0.9	0.5 to 0.1	S(p) grows from ~3 to ~19. Clusters begin to coalesce noticeably.
Near critical	0.9 to 1.0	0.1 to 0.0	S(p) diverges rapidly. Power-law growth $S \sim (p_c - p)^{-1}$ dominates.

The divergence of $S(p)$ as $p \rightarrow p_c$ is a **hallmark of critical phenomena**. It signals that clusters grow without bound as p approaches the critical point, reflecting the imminent formation of an infinite spanning cluster exactly at $p_c = 1$. The critical exponent $\gamma = 1$ is characteristic of the 1D percolation universality class.

Conclusions

Quantity	Formula	Key Behaviour
Spanning Probability	$\Pi(p; L) = p^L$	Step function at $p = 1$ as $L \rightarrow \infty$; $\Pi \rightarrow 0$ for all $p < 1$.
Critical Probability	$p_c = 1$	No spanning possible at any finite p ; unique to 1D.
Avg. Cluster Size	$S(p) = (1+p)/(1-p)$	Diverges as $p \rightarrow 1$: $S \sim (p_c - p)^{-1}$.
Critical Exponent	$\gamma = 1$	Power-law divergence; characteristic of 1D universality class.

Note on dimensionality: The result $p_c = 1$ is a purely 1D phenomenon. In higher dimensions the lattice topology provides alternative pathways around blocked sites. For the 2D square lattice $p_c \approx 0.5927$; for the 3D simple-cubic lattice $p_c \approx 0.3116$. Only in 1D does the absence of any bypass force p_c all the way to 1, making a genuine phase transition impossible at any finite p .